

Tongji Calculus, 7th Edition · Chapter 5

Definite Integrals

Exercise Workbook

100 classic-method adaptations and high-quality synthesis problems

How to Use This Workbook

The material progresses from basic to challenging. IDs Q001–Q100 match across both languages and both document types. Every formula comes from explicit LaTeX source and is compiled by the XeTeX mathematics engine.

Open-text method adaptations

20 / 100

Classic-method variants

50 / 100

Original synthesis / diagnosis

30 / 100

Source-lineage policy

Problems are redesigned around classic methods found in open textbooks, public courses, and standard calculus teaching. Copyrighted books are used only for scope and method alignment; their wording and solutions are not copied. See SOURCES.md.

1. On the first attempt, use only the exercise workbook and show complete reasoning.
2. During correction, record the cause of each error instead of copying the final answer.
3. Redo errors after 48 hours and interleave topics one week later.

- Basic
- Single choice
- Foundation
- Open-text method adaptation

Q001 · Recognizing a Riemann Sum on Equal Subintervals

Partition $[0, 2]$ into n equal pieces and use right endpoints. Which expression is a Riemann sum for $\int_0^2 (1 + x^2) dx$?

- A. $\sum_{k=1}^n \left[1 + \left(\frac{k}{n} \right)^2 \right] \frac{1}{n}$
- B. $\sum_{k=1}^n \left[1 + \left(\frac{2k}{n} \right)^2 \right] \frac{2}{n}$
- C. $\sum_{k=1}^n \left[1 + \left(\frac{2k}{n} \right)^2 \right]$
- D. $\sum_{k=0}^n \left[1 + \left(\frac{2k}{n} \right)^2 \right] \frac{2}{n}$

Workspace · add pages as needed

- Basic
- Single choice
- Foundation
- Open-text method adaptation

Q002 · Distinguishing an Integral from Geometric Area

If a continuous function satisfies $f(x) \leq 0$ on $[a, b]$, which statement is correct?

- A. $\int_a^b f(x) \, dx \geq 0$
- B. The geometric area equals $\int_a^b f(x) \, dx$.
- C. The geometric area equals $-\int_a^b f(x) \, dx$.
- D. The definite integral must equal 0.

Workspace · add pages as needed

- Basic
- Single choice
- Foundation
- Open-text method adaptation

Q003 · Reversing the Limits of Integration

Given $\int_1^4 f(x) \, dx = 7$, what is $\int_4^1 f(x) \, dx$?

- A. -7
- B. 0
- C. 7
- D. 14

Workspace · add pages as needed

- Basic
- Single choice
- Foundation
- Open-text method adaptation

Q004 · Bounding a Definite Integral

If f is integrable on $[1, 4]$ and $2 \leq f(x) \leq 5$ for every $x \in [1, 4]$, which statement must hold?

- A. $2 \leq \int_1^4 f(x) \, dx \leq 5$
- B. $3 \leq \int_1^4 f(x) \, dx \leq 12$
- C. $5 \leq \int_1^4 f(x) \, dx \leq 20$
- D. $6 \leq \int_1^4 f(x) \, dx \leq 15$

Workspace · add pages as needed

- Basic
- Multiple choice
- Foundation
- Open-text method adaptation

Q005 · Selecting Valid Integral Identities

Assume all displayed integrals exist. Which identities hold for every integrand?

- A. $\int_a^b \lambda f(x) \, dx = \lambda \int_a^b f(x) \, dx$
- B. $\int_a^b f(x)g(x) \, dx = \left(\int_a^b f(x) \, dx\right)\left(\int_a^b g(x) \, dx\right)$
- C. $\int_a^b f(x) \, dx = -\int_b^a f(x) \, dx$
- D. $\int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx$

Workspace · add pages as needed

- Basic
- Multiple choice
- Foundation
- Open-text method adaptation

Q006 · Recognizing Integrability and Order Consequences

On a closed interval $[a, b]$, which statements are correct?

- A. Every continuous function is Riemann integrable.
- B. Every monotone function is Riemann integrable.
- C. Equal integrals of two continuous functions imply pointwise equality.
- D. If $f(x) \geq 0$, then $\int_a^b f(x) \, dx \geq 0$.

Workspace · add pages as needed

Open-text method adaptation

Determine whether the statement is true and justify: if f is integrable on $[a, b]$, then $\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$.

Determine whether the statement is true and justify: if f is integrable on $[a, b]$, then $\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$.

Basic

True/false with justification

Foundation

Open-text method adaptation

Q008 · Inferring Equality of Functions from Equal Integrals

Determine whether the statement is true and justify: if continuous functions f, g satisfy $\int_0^1 f(x) \, dx = \int_0^1 g(x) \, dx$, then $f(x) = g(x)$ throughout $[0, 1]$.

Workspace · add pages as needed

- Basic
- True/false with justification
- Foundation
- Open-text method adaptation

Q009 · Symmetric Integral of an Odd Function

Determine whether the statement is true and justify: if f is an integrable odd function on $[-a, a]$, then $\int_{-a}^a f(x) \, dx = 0$.

Workspace · add pages as needed

Open-text method adaptation

For every integrable function f , $\int_a^a f(x) dx = \underline{\hspace{1cm}}$.

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Classic-method variant

The average value of $f(x) = 7$ on $[-2, 3]$ is ____.

- Basic
- Fill in the blank
- Foundation
- Classic-method variant

Q012 · Writing a Right-Endpoint Sum for a Square

Partition $[0, 1]$ into n equal pieces and use right endpoints. The Riemann sum for $f(x) = x^2$ is ____.

Workspace · add pages as needed

- Basic
- Calculation
- Foundation
- Classic-method variant

Q013 · Converting a Limit into a Definite Integral

Evaluate $\lim_{n \rightarrow \infty} \frac{27}{n^3} \sum_{k=1}^n k^2$ by first identifying it as a definite integral.

Workspace · add pages as needed

- Basic
- Calculation
- Foundation
- Classic-method variant

Q014 · Evaluating an Absolute-Value Integral Piecewise

Evaluate $\int_0^3 |x - 1| \, dx$.

Workspace · add pages as needed

- Basic
- Calculation
- Foundation
- Classic-method variant

Q015 · Combining Integrals over Adjacent Intervals

Let $a < b < c$, with $\int_a^b f(x) \, dx = 4$ and $\int_b^c f(x) \, dx = -1$. Find $\int_c^a f(x) \, dx$.

Workspace · add pages as needed

- Basic
- Calculation
- Foundation
- Classic-method variant

Q016 · Using Pointwise Bounds to Give the Sharp Guaranteed Range

Suppose f is integrable on $[-2, 2]$ and $-1 \leq f(x) \leq 3$. Find the range for $\int_{-2}^2 f(x) \, dx$ guaranteed by this information alone, and show that both bounds are attainable.

Workspace · add pages as needed

- Basic
- Calculation
- Foundation
- Classic-method variant

Q017 · Using Symmetry before Evaluation

Evaluate $\int_{-2}^2 (x^3 + 2x^2) \, dx$ and identify the part that vanishes.

Workspace · add pages as needed

Basic

Proof

Foundation

Original synthesis / diagnosis

Q018 · Proving Interval Additivity from Riemann Sums

Let f be continuous on $[a, b]$, with $a < c < b$. Starting from Riemann sums, prove $\int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx$.

Workspace · add pages as needed

- Basic
- Proof
- Foundation
- Original synthesis / diagnosis

Q019 · Proving the Integral Triangle Inequality

Let f be continuous on $[a, b]$. Prove $\left| \int_a^b f(x) \, dx \right| \leq \int_a^b |f(x)| \, dx$, and state a sufficient condition for equality.

Workspace · add pages as needed

- Basic
- Parameter / synthesis / application
- Foundation
- Original synthesis / diagnosis

Q020 · From a Nonlinear Sum to an Exact Integral

Find $L = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \sqrt{1 + \left(\frac{k}{n}\right)^2}$, identify the corresponding definite integral, and evaluate it exactly.

Workspace · add pages as needed

- Basic
- Single choice
- Foundation
- Classic-method variant

Q021 · Differentiating a Basic Accumulation Function

Let $F(x) = \int_0^x (1 + t^4) dt$. Which expression equals $F'(x)$?

- A. $1 + t^4$
- B. $4x^3$
- C. $1 + x^4$
- D. $x + x^5$

Workspace · add pages as needed

Basic

Single choice

Foundation

Classic-method variant

Q022 · A Variable Lower Limit with a Composite Endpoint

Let $H(x) = \int_{x^2}^3 e^{t^2} dt$. Which expression equals $H'(x)$?

- A. $2xe^{x^2}$
- B. $-2xe^{x^4}$
- C. e^{x^4}
- D. $-e^{x^2}$

Workspace · add pages as needed

- Standard
- Single choice
- Methods
- Classic-method variant

Q023 · Evaluating with the Newton-Leibniz Formula

Which value equals $\int_0^1 3x^2 \, dx$?

- A. 1
- B. $\frac{1}{3}$
- C. 3
- D. 0

Workspace · add pages as needed

- Standard
- Single choice
- Methods
- Classic-method variant

Q024 · Recognizing the Newton-Leibniz Formula

If $F'(x) = f(x)$ and f is continuous on $[a, b]$, which statement is correct?

- A. $\int_a^b f(x) \, dx = F(a) - F(b)$
- B. $\int_a^b f(x) \, dx = F(b)$
- C. $\int_a^b f(x) \, dx = f(b) - f(a)$
- D. $\int_a^b f(x) \, dx = F(b) - F(a)$

Workspace · add pages as needed

- Standard
- Multiple choice
- Methods
- Classic-method variant

Q025 · Necessary Properties of an Accumulation Function

Let f be continuous on $[a, b]$ and $F(x) = \int_a^x f(t) \, dt$. Which conclusions must hold?

- A. For $x \in (a, b)$, $F'(x) = f(x)$
- B. $F(a) = 0$
- C. $F''(x)$ must exist throughout the interval.
- D. $F(b) = \int_a^b f(t) \, dt$

Workspace · add pages as needed

- Standard
- Multiple choice
- Methods
- Classic-method variant

Q026 · Rules for an Integral with Two Variable Limits

Let f be continuous, let u, v be differentiable, and define $H(x) = \int_{u(x)}^{v(x)} f(t) \, dt$. Which conclusions are correct?

- A. $H'(x) = f(v(x))v'(x) - f(u(x))u'(x)$
- B. If $u(x) = v(x)$, then $H(x) = 0$.
- C. Interchanging u, v leaves H unchanged.
- D. If u is constant and $v(x) = x$, then $H'(x) = f(x)$.

Workspace · add pages as needed

Classic-method variant

Determine whether the statement is true and justify: if f is continuous on $[a, b]$ and $F(x) = \int_a^x f(t) dt$, then $F'(x) = f(x)$ for every $x \in (a, b)$.

Workspace · add pages as needed

Standard

True/false with justification

Methods

Classic-method variant

Q028 · One Zero Accumulation Value Does Not Determine an Endpoint Value

Determine whether the statement is true and justify: if a continuous function satisfies $\int_0^1 f(t) \, dt = 0$, then necessarily $f(1) = 0$.

Workspace · add pages as needed

Classic-method variant

Fill in the blank: $\int_1^e \frac{1}{x} dx = \underline{\hspace{2cm}}$.

- Standard
- Fill in the blank
- Methods
- Classic-method variant

Q030 · Differentiation with Trigonometric Endpoints

Let $F(x) = \int_{\sin x}^{\cos x} e^{t^2} dt$. Then $F'(x) = \underline{\hspace{1cm}}$.

Workspace · add pages as needed

Classic-method variant

Let $G(x) = x \int_0^x t^2 dt$. Then $G'(x) = \underline{\hspace{2cm}}$.

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Classic-method variant

Fill in the blank: $\int_0^{\frac{\pi}{2}} \sin x \, dx = \underline{\hspace{1cm}}$.

Fill in the blank: $\int_0^{\frac{\pi}{2}} \sin x \, dx = \underline{\hspace{1cm}}$.

Classic-method variant

Let $G(x) = \int_1^{x^2} \ln t \, dt$. Then $G'(1) = \underline{\hspace{1cm}}$.

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- Standard
- Calculation
- Methods
- Classic-method variant

Q034 · A Definite Integral of a Linear Function

Evaluate $\int_{-1}^2 (2x + 3) \, dx$.

Workspace · add pages as needed

- Standard
- Calculation
- Methods
- Classic-method variant

Q035 · A Definite Integral with the Arctangent Antiderivative

Evaluate $\int_0^1 \frac{1}{1+x^2} dx$.

Workspace · add pages as needed

- Standard
- Calculation
- Methods
- Classic-method variant

Q036 · An Exponential Integral over a Finite Interval

Evaluate $\int_0^{\ln 2} e^{2x} dx$.

Workspace · add pages as needed

- Standard
- Calculation
- Methods
- Classic-method variant

Q037 · Integrating the Square of Cosine over a Half Period

Evaluate $\int_0^\pi \cos^2 x \, dx$.

Workspace · add pages as needed

- Standard
- Calculation
- Methods
- Classic-method variant

Q038 · Differentiating Two Composite Variable Endpoints

Let $F(x) = \int_x^{x^2} \ln(1 + t^2) dt$. Find $F'(x)$.

Workspace · add pages as needed

- Standard
- Calculation
- Methods
- Original synthesis / diagnosis

Q039 · An Integral Function with a Parameter-Dependent Kernel

Let f be continuous and $G(x) = \int_0^x (x - t)f(t) \, dt$. Find $G'(x)$ after first separating the explicit occurrence of x in the integrand.

Workspace · add pages as needed

- Standard
- Calculation
- Methods
- Original synthesis / diagnosis

Q040 · Recovering an Integrand from Its Accumulation

A continuous function f satisfies $\int_0^x f(t) \, dt = x^2 e^x$. Find $f(x)$.

Workspace · add pages as needed

- Standard
- Calculation
- Methods
- Original synthesis / diagnosis

Q041 · A Second Derivative of an Accumulation Function

Let $F(x) = \int_0^{x^2} \cos t \, dt$. Find $F''(x)$.

Workspace · add pages as needed

- Standard
- Proof
- Methods
- Original synthesis / diagnosis

Q042 · Proving the Fundamental Theorem with the Integral Mean Value Theorem

Let f be continuous on $[a, b]$ and $F(x) = \int_a^x f(t) dt$. Prove that $F'(x) = f(x)$ for $x \in (a, b)$.

Workspace · add pages as needed

Standard

Proof

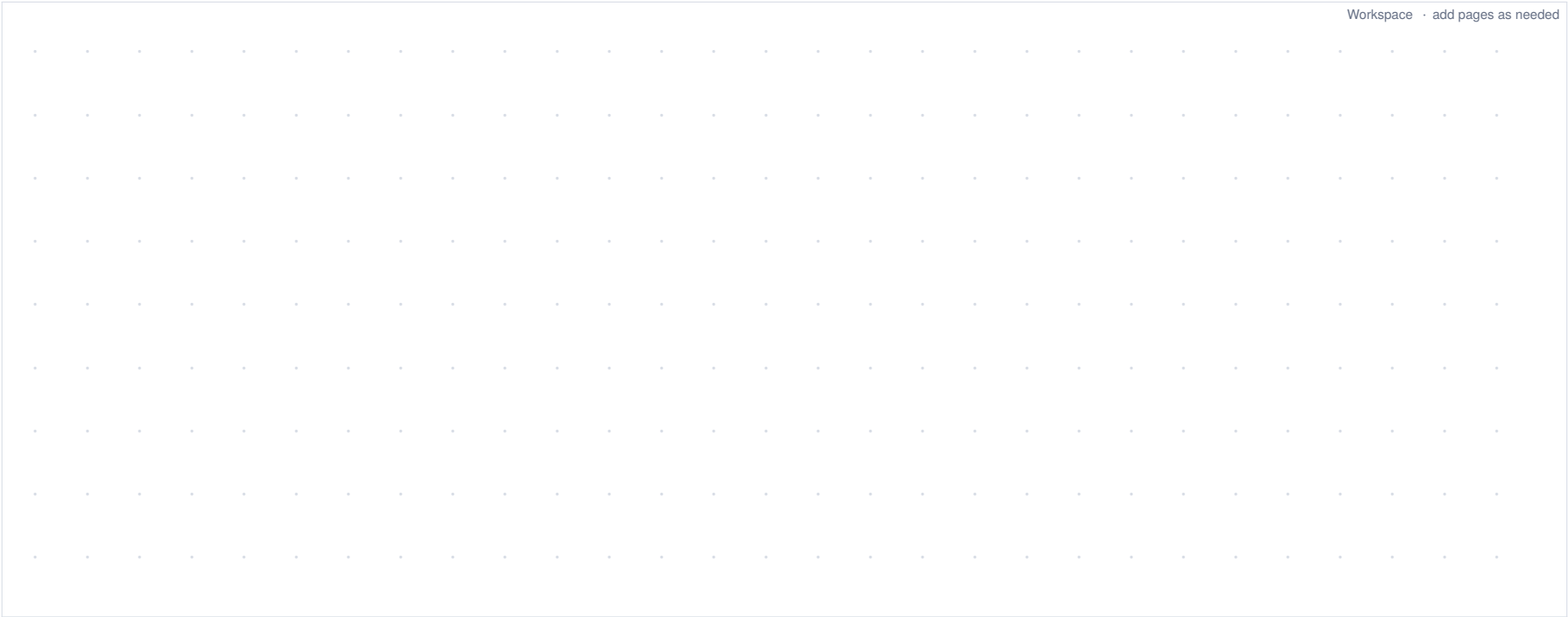
Methods

Original synthesis / diagnosis

Q043 · A Zero Accumulation Function Forces a Zero Integrand

Let f be continuous on $[a, b]$, and suppose $\int_a^x f(t) \, dt = 0$ for every $x \in [a, b]$. Prove that $f(x) = 0$ throughout $[a, b]$.

Workspace · add pages as needed



- Standard
- Error diagnosis
- Methods
- Original synthesis / diagnosis

Q044 · Diagnosing Missing Factors in Two-Limit Differentiation

For $H(x) = \int_x^{x^2} f(t) dt$, a student writes $H'(x) = f(x^2) - f(x)$. Identify the error and give the correct result when f is continuous.

Workspace · add pages as needed

- Standard
- Calculation
- Methods
- Classic-method variant

Q045 · Substitution in a Logarithmic Definite Integral

Use substitution to evaluate $\int_0^1 \frac{2x}{1+x^2} dx$, transforming both limits.

Workspace · add pages as needed

- Advanced
- Calculation
- Methods
- Original synthesis / diagnosis

Q046 · A Trigonometric Substitution with Reversed New Limits

Use substitution to evaluate $\int_0^{\frac{\pi}{2}} \sin x \cos^3 x \, dx$.

Workspace · add pages as needed

- Advanced
- Calculation
- Methods
- Classic-method variant

Q047 · Integration by Parts for a Polynomial Times an Exponential

Use integration by parts to evaluate $\int_0^1 x e^x dx$.

Workspace · add pages as needed

- Advanced
- Calculation
- Methods
- Original synthesis / diagnosis

Q048 · Integration by Parts with a Trigonometric Factor

Evaluate $\int_0^{\frac{\pi}{2}} x \sin x \, dx$.

Workspace · add pages as needed

- Advanced
- Proof
- Methods
- Original synthesis / diagnosis

Q049 · An Interval-Reflection Substitution Identity

Let f be continuous on $[0, a]$ with $a > 0$. Prove $\int_0^a f(x) \, dx = \int_0^a f(a - x) \, dx$.

Workspace · add pages as needed

Advanced

Proof

Methods

Original synthesis / diagnosis

Q050 · A Symmetric Ratio Integral via Complementary Angles

Let f be continuous and strictly positive on $[0, 1]$. Prove $\int_0^{\frac{\pi}{2}} \frac{f(\sin x)}{f(\sin x)+f(\cos x)} dx = \frac{\pi}{4}$.

Workspace · add pages as needed

Open-text method adaptation

Evaluate $\int_0^1 x(1+x^2)^2 dx = \underline{\hspace{1cm}}$.

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- Standard
- Single choice
- Synthesis
- Open-text method adaptation

Q052 · A Definite Integral of an Exponential Composite

Which value equals $\int_0^1 x e^{x^2} dx$?

- A. $e - 1$
- B. $\frac{e-1}{2}$
- C. $\frac{e}{2}$
- D. $\frac{1}{2}$

Workspace · add pages as needed

- Standard
- Multiple choice
- Synthesis
- Original synthesis / diagnosis

Q053 · Consistency of Substitution and Endpoints

Which substitution results are fully correct? Assume every displayed f is continuous.

A. $\int_0^1 2xf(x^2) \, dx = \int_0^1 f(u) \, du$

B. $\int_1^e \frac{f(\ln x)}{x} \, dx = \int_0^1 f(u) \, du$

C. With $u = \cos x$, $\int_0^{\frac{\pi}{2}} \sin x \, dx = \int_1^0 du$

D. If $a > 0$, $\int_0^a f\left(\frac{x}{a}\right) \, dx = a \int_0^1 f(u) \, du$

Workspace · add pages as needed

Original synthesis / diagnosis

True or false: if $u = \varphi(x)$ is strictly decreasing on $[a, b]$, one may always put the smaller transformed endpoint below the larger one without adding a minus sign.

Workspace · add pages as needed

- Standard
- Calculation
- Synthesis
- Open-text method adaptation

Q055 · Substitution for an Odd Power of Sine

Evaluate $\int_0^{\frac{\pi}{2}} \sin^3 x \cos^2 x \, dx$.

Workspace · add pages as needed

- Standard
- Calculation
- Synthesis
- Open-text method adaptation

Q056 · A Reverse-Chain Structure in a Radical

Evaluate $\int_0^{\frac{\sqrt{3}}{2}} \frac{x}{\sqrt{1-x^2}} dx.$

Workspace · add pages as needed

- Standard
- Fill in the blank
- Synthesis
- Classic-method variant

Q057 · Endpoint Substitution in an Exponential Fraction

Evaluate $\int_0^{\ln 2} \frac{e^x}{1 + e^x} dx = \underline{\hspace{1cm}}.$

Workspace · add pages as needed

- Standard
- Single choice
- Synthesis
- Open-text method adaptation

Q058 · The Boundary Term in Integration by Parts

Let u, v be continuously differentiable on $[a, b]$. Which formula for definite integration by parts is correct?

- A. $\int_a^b uv' \, dx = uv - \int_a^b u'v \, dx$
- B. $\int_a^b uv' \, dx = [uv]_a^b + \int_a^b u'v \, dx$
- C. $\int_a^b uv' \, dx = [uv]_a^b - \int_a^b u'v \, dx$
- D. $\int_a^b uv' \, dx = \int_a^b u'v \, dx$

Workspace · add pages as needed

- Standard
- Calculation
- Synthesis
- Classic-method variant

Q059 · Integration by Parts with a Logarithmic Factor

Evaluate $\int_0^1 x \ln(1+x) \, dx$.

Workspace · add pages as needed

Standard

Proof

Synthesis

Original synthesis / diagnosis

Q060 · A Chain-Rule Proof of Definite Substitution

Let f be continuous on an interval containing $\varphi([\alpha, \beta])$, and let φ be continuously differentiable on $[\alpha, \beta]$. Prove $\int_{\alpha}^{\beta} f(\varphi(t))\varphi'(t) dt = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(u) du$, and explain why monotonicity of φ is unnecessary.

Workspace · add pages as needed

- Advanced
- Multiple choice
- Synthesis
- Original synthesis / diagnosis

Q061 · Substitution Identities with a General Function

Let f be continuous on the relevant intervals and let $a > 0$. Which identities hold?

- A. $\int_0^1 x f(x^2) dx = \frac{1}{2} \int_0^1 f(u) du$
- B. $\int_{-1}^1 x f(x^2) dx = \int_0^1 f(u) du$
- C. $\int_{-1}^1 f(x^2) dx = \int_0^1 \frac{f(u)}{\sqrt{u}} du$
- D. $\int_0^a f\left(\frac{x}{a}\right) dx = a \int_0^1 f(u) du$

Workspace · add pages as needed

- Advanced
- True/false with justification
- Synthesis
- Original synthesis / diagnosis

Q062 · A Nonmonotone Map Can Still Satisfy the Substitution Formula

True or false: if f is continuous, then even though $u = x^2$ is not monotone on $[-1, 1]$, the definite-integral substitution formula still gives $\int_{-1}^1 xf(x^2) dx = \frac{1}{2} \int_1^1 f(u) du = 0$ directly.

Workspace · add pages as needed

- Advanced
- Calculation
- Synthesis
- Classic-method variant

Q063 · Rational Decomposition after Substitution

Evaluate $\int_0^1 \frac{x^3}{(1+x^2)^2} dx$.

Workspace · add pages as needed

- Advanced
- Calculation
- Synthesis
- Classic-method variant

Q064 · A Linear Factor Times Cosine

Evaluate $\int_0^{\frac{\pi}{2}} x \cos x \, dx$.

Workspace · add pages as needed

- Advanced
- Proof
- Synthesis
- Original synthesis / diagnosis

Q065 · A Weighted Integral with Midpoint Symmetry

Let $a > 0$ and let f be continuous on $[0, a]$ with $f(a - x) = f(x)$. Prove $\int_0^a x f(x) \, dx = \frac{a}{2} \int_0^a f(x) \, dx$.

Workspace · add pages as needed

- Advanced
- Parameter / synthesis / application
- Synthesis
- Original synthesis / diagnosis

Q066 · A Parameterized Semicircle-Radical Integral

Let $a > 0$. Evaluate $I(a) = \int_0^a x\sqrt{a^2 - x^2} \, dx$ and identify its scaling power in a .

Workspace · add pages as needed

- Advanced
- Error diagnosis
- Synthesis
- Original synthesis / diagnosis

Q067 · Diagnosing a Non-injective Substitution

A student evaluates $J = \int_{-1}^1 |x|e^{x^2} dx$ by setting $u = x^2$ on the whole interval and writes $J = \frac{1}{2} \int_1^1 e^u du = 0$ because both transformed endpoints are

1. Identify the error and find the correct value.

Workspace · add pages as needed

- Advanced
- Calculation
- Synthesis
- Classic-method variant

Q068 · A Definite Integral of Arctangent

Evaluate $\int_0^1 \arctan x \, dx$.

Workspace · add pages as needed

- Advanced
- Proof
- Synthesis
- Classic-method variant

Q069 · The Wallis Integral Recurrence

For an integer $n \geq 2$, let $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$. Prove $I_n = \frac{n-1}{n} I_{n-2}$ and state I_0, I_1 .

Workspace · add pages as needed

- Advanced
- Proof
- Synthesis
- Original synthesis / diagnosis

Q070 · A Weighted-Derivative Identity from Integration by Parts

Let $a > 0$ and let f be continuously differentiable on $[0, a]$. Prove $\int_0^a x f'(x) \, dx = a f(a) - \int_0^a f(x) \, dx$.

Workspace · add pages as needed

Open-text method adaptation

For real p , determine whether the following statement is true: the improper integral $\int_1^{+\infty} \frac{dx}{x^p}$ converges if and only if $p > 1$.

- Advanced
- Calculation
- Synthesis
- Open-text method adaptation

Q072 · A Basic Improper Integral on an Infinite Interval

Evaluate $\int_1^{+\infty} \frac{dx}{x^2}$.

Workspace · add pages as needed

- Advanced
- Calculation
- Synthesis
- Open-text method adaptation

Q073 · An Integrable Power Singularity at an Endpoint

Evaluate $\int_0^1 \frac{dx}{\sqrt{x}}$.

Workspace · add pages as needed

- Advanced
- Proof
- Synthesis
- Classic-method variant

Q074 · Proving Logarithmic Divergence by Comparison

Prove that $\int_1^{+\infty} \frac{dx}{x+1}$ diverges.

Workspace · add pages as needed

- Advanced
- Parameter / synthesis / application
- Synthesis
- Original synthesis / diagnosis

Q075 · A Principal Value Exists but the Ordinary Integral Diverges

Discuss the ordinary improper integral and the Cauchy principal value of $\int_{-1}^1 \frac{dx}{x}$.

Workspace · add pages as needed

- Advanced
- Calculation
- Synthesis
- Classic-method variant

Q076 · A Cube-Root Endpoint Singularity

Evaluate $\int_0^1 \frac{dx}{\sqrt[3]{x}}.$

Workspace · add pages as needed

- Advanced
- Calculation
- Synthesis
- Classic-method variant

Q077 · The Parameter Sign in Exponential Decay

For real a , determine convergence and evaluate $I(a) = \int_0^{+\infty} e^{-ax} dx$.

Workspace · add pages as needed

- Advanced
- Calculation
- Synthesis
- Classic-method variant

Q078 · Cancellation of Logarithms at Infinity

Evaluate $\int_1^{+\infty} \frac{dx}{x(x+1)}.$

Workspace · add pages as needed

Advanced

Proof

Synthesis

Original synthesis / diagnosis

Q079 · The Cauchy Tail Criterion for an Improper Integral

Let f be continuous on every finite interval $[a, R]$. Prove that $\int_a^{+\infty} f(x) dx$ converges if and only if, for every $\varepsilon > 0$, there exists $M > a$ such that all $B > A \geq M$ satisfy $\left| \int_A^B f(x) dx \right| < \varepsilon$.

Workspace · add pages as needed

- Advanced
- Parameter / synthesis / application
- Synthesis
- Original synthesis / diagnosis

Q080 · Independent Splitting of a Two-sided Infinite Integral

Using the ordinary improper-integral definition, evaluate $\int_{-\infty}^{+\infty} \frac{dx}{1+x^2}$.

Workspace · add pages as needed

- Advanced
- Error diagnosis
- Synthesis
- Original synthesis / diagnosis

Q081 · Misusing Symmetry as Ordinary Convergence

A student argues that $\frac{x}{1+x^2}$ is odd and therefore $\int_{-\infty}^{+\infty} \frac{x}{1+x^2} dx = 0$. Diagnose the error and determine the ordinary integral and symmetric principal value.

Workspace · add pages as needed

- Advanced
- Calculation
- Synthesis
- Classic-method variant

Q082 · The Integrable Logarithmic Singularity

Evaluate $\int_0^1 \ln x \, dx$.

Workspace · add pages as needed

Hard

Calculation

Synthesis

Classic-method variant

Q083 · A Substitution with Both Zero and Infinite Endpoints

Evaluate $\int_0^{+\infty} \frac{dx}{(1+x)\sqrt{x}}.$

Workspace · add pages as needed

Hard

Proof

Synthesis

Classic-method variant

Q084 · The Comparison Test for Nonnegative Functions

Assume $0 \leq f(x) \leq g(x)$ for all $x \geq a$, with both functions integrable on finite intervals. Prove: if $\int_a^{+\infty} g(x) \, dx$ converges, then $\int_a^{+\infty} f(x) \, dx$ converges; if $\int_a^{+\infty} f(x) \, dx$ diverges, then $\int_a^{+\infty} g(x) \, dx$ diverges.

Workspace · add pages as needed

Hard

Parameter / synthesis / application

Synthesis

Original synthesis / diagnosis

Q085 · An Infinite Integral with a Parameterized Positive Quadratic

Let $a > 0$. Evaluate $I(a) = \int_0^{+\infty} \frac{dx}{x^2 + 2ax + 2a^2}$.

Workspace · add pages as needed

Hard

Error diagnosis

Synthesis

Original synthesis / diagnosis

Q086 · Applying Newton-Leibniz across a Second-order Singularity

A student writes $\int_0^2 \frac{dx}{(x-1)^2} = -\frac{1}{x-1} \Big|_0^2 = -2$. Identify all errors and give the correct conclusion.

Workspace · add pages as needed

Hard

Calculation

Synthesis

Classic-method variant

Q087 · Algebraic Decay and Substitution

Evaluate $\int_0^{+\infty} \frac{x}{(1+x^2)^{\frac{3}{2}}} dx.$

Workspace · add pages as needed

Hard

Proof

Synthesis

Classic-method variant

Q088 · Reciprocal Substitution between Infinity and Zero

Let f be continuous on $[1, +\infty)$. Prove that whenever either side converges, $\int_1^{+\infty} f(x) \, dx = \int_0^1 \frac{f\left(\frac{1}{t}\right)}{t^2} \, dt$, and show that the two sides converge or diverge together.

Workspace · add pages as needed

Hard

Calculation

Synthesis

Open-text method adaptation

Q089 · Complete Parameter Classification for an Endpoint Power Integral

For real p , determine convergence and evaluate $I(p) = \int_0^1 \frac{dx}{x^p}$.

Workspace · add pages as needed

Hard

Proof

Synthesis

Classic-method variant

Q090 · The Limit Comparison Test

Let positive functions f, g be integrable on every finite interval $[a, R]$, and suppose $\lim_{x \rightarrow +\infty} \frac{f(x)}{g(x)} = L$ with $0 < L < +\infty$. Prove that $\int_a^{+\infty} f(x) \, dx$ and $\int_a^{+\infty} g(x) \, dx$ have the same convergence behavior.

Workspace · add pages as needed

Hard

Calculation

Synthesis

Classic-method variant

Q091 · A Logarithmically Corrected Power Integral

For real p , determine convergence and evaluate $J(p) = \int_e^{+\infty} \frac{dx}{x(\ln x)^p}$.

Workspace · add pages as needed

Hard

Proof

Synthesis

Original synthesis / diagnosis

Q092 · Conditional Convergence of an Oscillatory Integral

Prove that $\int_1^{+\infty} \frac{\sin x}{x} dx$ converges but not absolutely.

Workspace · add pages as needed

Hard

Parameter / synthesis / application

Synthesis

Open-text method adaptation

Q093 · The Convergence Parameter of the Gamma Integral

For real s , determine convergence of $\Gamma(s) = \int_0^{+\infty} x^{s-1} e^{-x} dx$.

Workspace · add pages as needed

Hard

Proof

Synthesis

Classic-method variant

Q094 · The Gamma Recurrence

Let $s > 0$. Prove $\Gamma(s + 1) = s\Gamma(s)$ and justify both boundary limits used in integration by parts.

Workspace · add pages as needed

Hard

Calculation

Synthesis

Classic-method variant

Q095 · Gamma Function and Factorials

Let n be a nonnegative integer. Use the definition and recurrence to find $\Gamma(n + 1)$.

Workspace · add pages as needed

Hard

Parameter / synthesis / application

Synthesis

Classic-method variant

Q096 · Scaling Substitution and a Gamma Value

Evaluate $\int_0^{+\infty} x^3 e^{-2x} dx.$

Workspace · add pages as needed

Challenge

Calculation

Challenge

Classic-method variant

Q097 · A Half-integer Gamma Value

Given $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$, evaluate $\Gamma\left(\frac{5}{2}\right)$.

Workspace · add pages as needed

Challenge

Calculation

Challenge

Classic-method variant

Q098 · A Factorial Integral with Exponential Scaling

Let $a > 0$ and let m be a nonnegative integer. Evaluate $I_m(a) = \int_0^{+\infty} x^m e^{-ax} \, dx$.

Workspace · add pages as needed

- Challenge
- Proof
- Challenge
- Classic-method variant

Q099 · Two-endpoint Convergence of a Beta-type Integral

For real α, β , prove that $\int_0^{+\infty} \frac{x^{\alpha-1}}{(1+x)^{\alpha+\beta}} dx$ converges if and only if $\alpha > 0$ and $\beta > 0$.

Workspace · add pages as needed

Challenge

Parameter / synthesis / application

Challenge

Original synthesis / diagnosis

Q100 · A Two-parameter Gamma Scaling Integral

For real parameters a, b , completely classify and evaluate $I(a, b) = \int_0^{+\infty} x^{a-1} e^{-bx} dx$.

Workspace · add pages as needed